



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech IV Semester - End Semester Examination - April / May -2025

Analog Integrated Circuits [ECM-2223]

Marks: 50

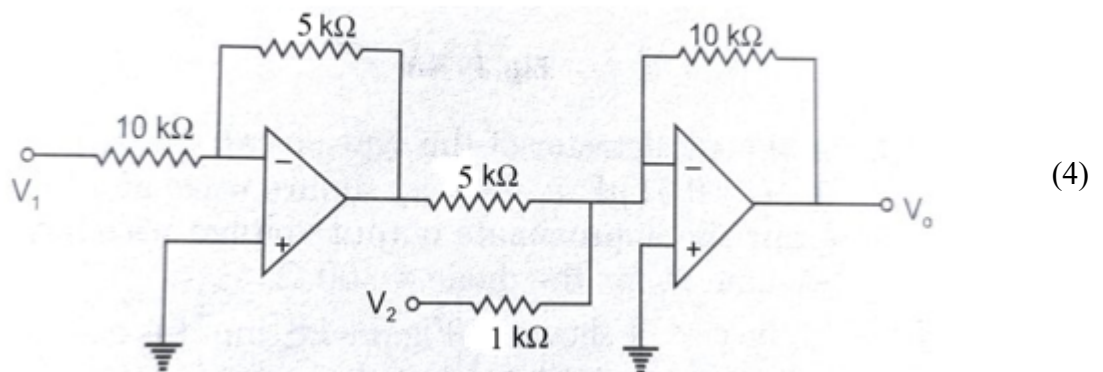
Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Determine the output voltage of the circuit given below if $v_1 = -0.5\text{V}$ and $v_2 = +0.3\text{V}$.

A)



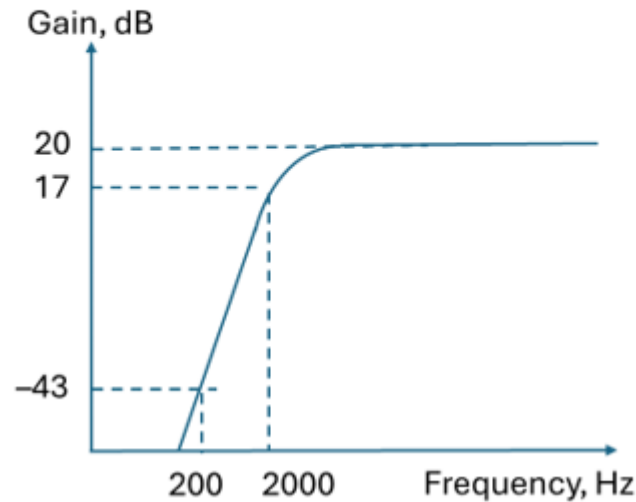
- B) In a voltage-follower circuit, suppose the input voltage is $5 \sin(2\pi ft)$, determine the maximum frequency “f” at which the output voltage waveform is undistorted, if the slew-rate of the op-amp is $1 \text{ V}/\mu\text{s}$.
- (3)

- C) (i) Using a circuit diagram, demonstrate how the output resistance of an op-amp can be measured practically. (ii) Suppose the output of op-amp is 2 V when $R_L = \infty$, and the output is 1.75 V when $R_L = 220 \Omega$, determine the output resistance of the op-amp.
- (3)

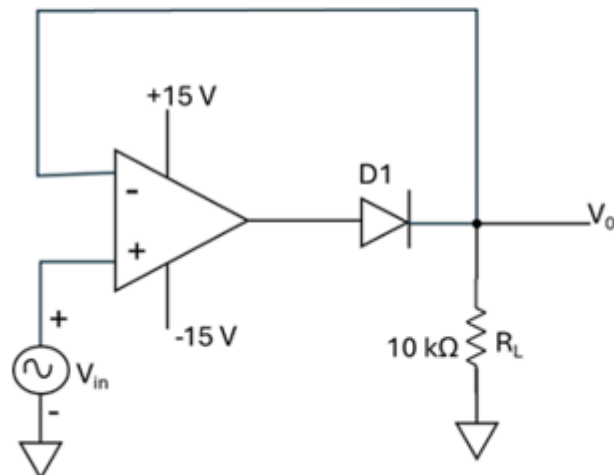
- 2) (4)

A)

Design a circuit using op-amps to obtain the frequency response shown in the figure below. Use $C = 0.047 \mu\text{F}$ and appropriate resistor values in your design.



- B) Identify the circuit given below and mention its drawback. Illustrate how this drawback is overcome with an improved circuit. Analyze the second circuit and draw the input and output waveforms if $v_{in} = 0.5 \sin(200\pi t)$.



(3)

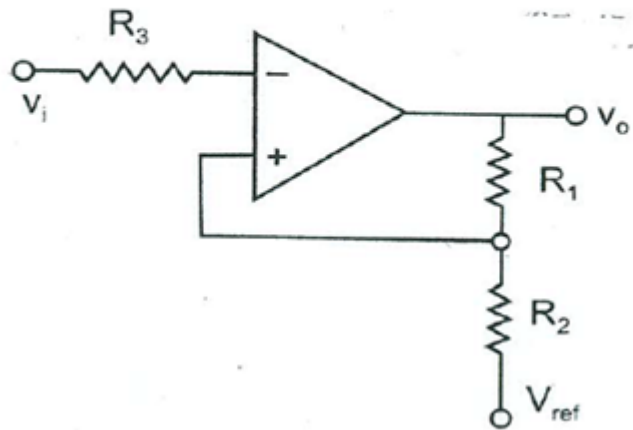
- C) Design a first order filter to pass the signals of frequencies between 500 Hz and 1 kHz with a gain of 20 dB and attenuate the signals outside this frequency range. Choose $C = 0.068 \mu\text{F}$.

(3)

3) (4)

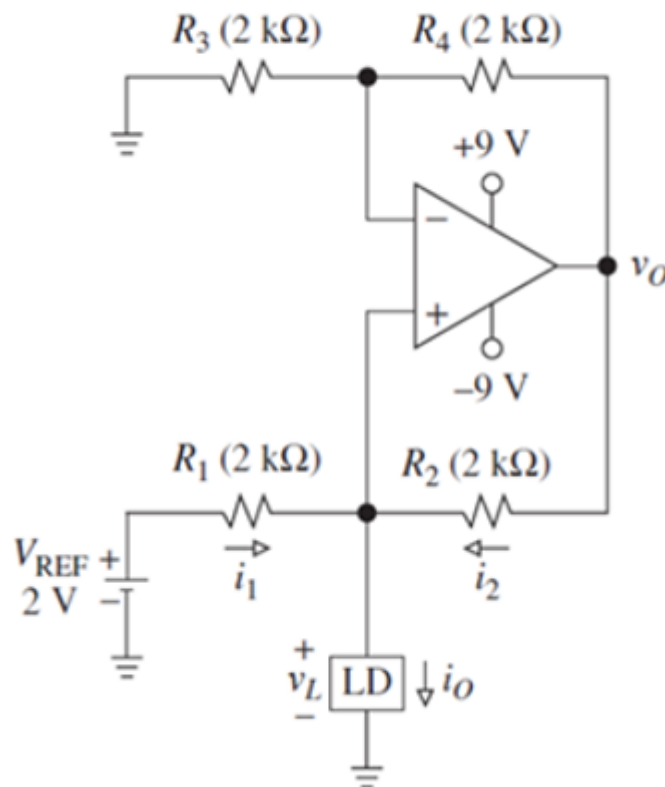
A)

Identify the circuit shown below. Suppose if $V_{ref} = 2\text{ V}$, $R_1 = 1\text{ k}\Omega$, $R_2 = 100\text{ }\Omega$, $R_3 = 10\text{ k}\Omega$ and $\pm V_{sat} = \pm 10\text{ V}$, draw the output waveform if input $v_i = 5 \sin(1000\pi t)$.



B) Using an op-amp and appropriate nonlinear component, show how a logarithmic relationship between input and output voltages can be achieved. Draw the circuit diagram and derive the mathematical expression for the output voltage. (3)

C) Analyze the given circuit and determine the currents i_o , i_1 , and i_2 for load resistances (i) $R_L = 1\text{ k}\Omega$, (ii) $R_L = 2\text{ k}\Omega$ and (iii) $R_L = 4\text{ k}\Omega$.



(3)

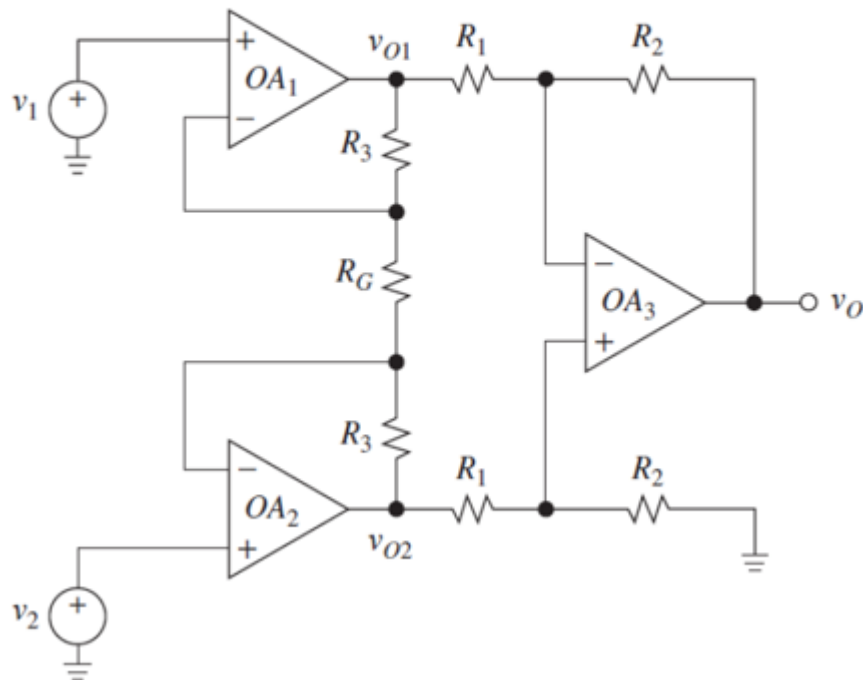
4) Design a monostable multivibrator using a 555-timer IC to produce a pulse of 100 ms width. Use $C = 1\text{ }\mu\text{F}$. Draw the circuit and the resulting waveforms. Mention two practical applications of the above circuit. (4)

A)

- B) With a neat diagram, explain the working of 8-bit flash ADC. (3)
- C) Draw the circuit of 4-bit R-2R DAC with $R = 22 \text{ k}\Omega$, $R_f = 47 \text{ k}\Omega$ and reference voltage of -5 V . Determine the output voltage if the digital input is (i) 1010 and (ii) 1111. (3)

- 5) With block schematics, explain how PLL can be used as (i) frequency multiplication/division and (ii) AM detection. (4)

- A)
- B) Compare the voltage regulator ICs LM317 and 7805. Explain the working of each. (3)
- C) For the circuit given below, determine the expression for output voltage. Suppose $R_1 = 10 \text{ k}\Omega$, $R_2 = 100 \text{ k}\Omega$, $R_3 = 50 \text{ k}\Omega$, $R_G = 1 \text{ k}\Omega$, $v_1 = -1 \text{ mV}$ and $v_2 = 5 \text{ mV}$, what will be the output voltage?



(3)

-----End-----